Darsh Thakkar

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Professional Summary

Software and ML engineer building reliable AI applications, LLM-powered systems, and robotics interfaces. Experienced in Python, C++, React, and backend development, with a focus on privacy, structured AI workflows, and production-oriented testing.

Experience

Undergraduate Research Assistant – People and Robots Lab, UW-Madison — ***Feb 2026 – Present* | Madison, WI, USA**

- Developed a Flask-based failure-handling pipeline that converts Temi robot failure events, including blocked obstacles, low battery, and location changes, into structured LLM-generated fallback behaviors.
- Built React components for a robot programming interface with real-time status visualization, user feedback collection, condition switching, and retry-state handling.
- Integrated and tested the interface on a physical Temi robot using Docker, WSL, Android tooling, and backend communication for scenario execution and fallback testing.

AI and Software Engineer Intern — LexLegis AI — ***May 2025 - Jul 2025* | Mumbai, India**

- Developed a fully offline AI legal assistant with chat, OCR, multilingual translation, and semantic search to ensure data privacy.
- Built cross-platform desktop features in C++, Python, and Qt/QML; integrated local LLM inference via llama.cpp and Llama Server.
- Implemented RAG architectures for grounded answer retrieval (RAG + BM25 + embeddings) across document collections.

Software Development Intern — Yodaplus — ***Jun 2024 - Aug 2024* | Mumbai, India**

- Reviewed SQL queries and backend behavior to identify bugs and propose fixes; collaborated with engineers to resolve issues.
- Wrote Pandas-based data cleaning/validation scripts to improve data quality for downstream analytics and APIs.
- Developed full-stack features using Python, REST APIs, PostgreSQL, and React.

Projects

ShadowOps — AI Deployment Preflight Platform | IBM July Wildcard Hackathon | GitHub

- Built and deployed a full-stack platform that reconstructs operational workflows from structured event logs, surfaces hidden human effort, and evaluates proposed AI automation before deployment.
- Designed a deterministic AI Tax model measuring five overhead categories, net savings, burden concentration, and deployment readiness; measured 20.5% overhead across a synthetic 46-event workflow.
- Integrated IBM Granite through watsonx.ai to generate grounded recommendations and safer hybrid workflows, adding PII redaction, structured-output validation, cached fallback, and 60 backend tests.

Voix — AI Voice Copilot for Windows (CalHacks 12.0) | GitHub

- Built a Windows voice copilot for controlling applications, generating documents, and coordinating multi-app workflows through natural-language commands.
- Integrated Faster-Whisper, Groq, Claude, and Python COM automation across Word, Excel, and Chrome.
- Developed a floating desktop interface with asyncio, Silero VAD, meeting transcription, and local action-item extraction.

Little Lemon Restaurant API — Meta Back-End Developer Capstone | GitHub

- Built a Django REST API with MySQL-backed models and endpoints for restaurant menu and table-booking operations.
- Implemented user registration and authentication, secured API endpoints, and validated functionality with unit and API tests.

Education

University of Wisconsin-Madison — B.S. Computer Science and Data Science (Expected May 2027)

GPA: 3.33 | Dean's List: Fall 2023, Fall 2024

Skills

Programming Languages: Python, C++, Java, JavaScript, TypeScript, SQL, R, C
AI/ML: PyTorch, scikit-learn, OpenCV, MediaPipe, NLP, CNNs, Reinforcement Learning
LLM Systems: IBM Granite, watsonx.ai, llama.cpp, RAG, embeddings, BM25, GGUF, structured outputs, prompt engineering, Tesseract OCR
Web/Backend: React, Django, Django REST Framework, FastAPI, Flask, REST APIs, MySQL, PostgreSQL, SQLite
Tools/Deployment: Docker, Render, Git, CMake, FFmpeg, pytest, Qt/QML

Certifications

Meta Back-End Developer Professional Certificate — Meta, 2026
Supervised Machine Learning — DeepLearning.AI / Stanford Online, 2025